

Pilot 01: Tooling & Infrastructure Plan

Volunteer-Built Agricultural AI Dataset
Ireland, 2026

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1. Introduction and Design Principles

This document specifies the tooling and infrastructure required to run Pilot 01: the volunteer-built agricultural AI dataset for Irish weed identification.

Context: The pilot tests whether non-experts can collect, annotate, and peer-review agricultural image data to a standard credible for downstream machine learning. The tooling must support this without itself becoming a bottleneck or distraction. We are testing human systems, not building production infrastructure.

Design principles for tool selection:

- **Cheap:** Total tooling budget target of €500–€1,500 for the 12-week pilot. Free tiers and open-source tools are preferred wherever they meet requirements.
- **Quick to deploy:** The full stack should be operational within 3-4 weeks of a small team starting work. No tool should require more than a day or two of configuration.
- **User-friendly for non-technical volunteers:** Students and adult volunteers will interact with data collection and annotation interfaces. These must work on mobile devices, require no software installation, and need minimal training.
- **Disposable:** Nothing in this stack is intended to become permanent infrastructure. We learn from this pilot and then design purpose-built tooling for future iterations. Avoid lock-in.
- **Observable:** The team needs visibility into data volumes, annotation progress, quality metrics, and volunteer activity without building custom dashboards from scratch.

2. Architecture Overview

The pilot's tooling breaks into six functional areas, each serving a distinct phase of the operational flow described in the pilot outline. The areas are: data collection and ingestion, image storage and management, annotation and labelling, quality assurance and gold-standard management, analysis and ML sanity-checking, and coordination and documentation.

Functional Area	Primary Tool	Alternatives Considered	Pilot Phase
Data collection & ingestion	KoboToolbox (free hosted)	Google Forms + Drive, ODK Collect	Steps C–D (Weeks 4–6)
Image storage & management	Google Drive (shared, free tier) or institutional Google Workspace	S3 bucket, Nextcloud	All phases
Annotation & labelling	Label Studio Community Edition (self-hosted)	CVAT, Labelbox free tier	Steps E–F (Weeks 6–10)
Quality assurance & gold standards	Label Studio + scripted analytics (Python)	Custom app, spreadsheet tracking	Steps E–G (Weeks 6–12)
Analysis & ML sanity check	Google Colab (free/Pro) or local machine	Kaggle Notebooks, university HPC	Steps F–G (Weeks 8–12)
Coordination & documentation	Google Workspace (Docs, Sheets, Meet)	Notion, Slack free tier	All phases

The remainder of this document specifies each area in detail: what the tool does, how it is configured, what the volunteer and team experience looks like, estimated costs, and known risks.

3. Data Collection and Ingestion

3.1 Recommended Tool: KoboToolbox

What it is: KoboToolbox is a free, open-source data collection platform originally built for humanitarian field research. It provides a mobile-friendly form interface (KoboCollect app or web browser), supports offline data capture, handles image attachments, and includes GPS and metadata capture out of the box. The hosted version at kf.kobotoolbox.org is free for non-commercial use.

Why this over Google Forms: Google Forms can accept image uploads, but it lacks offline capability, enforced metadata capture, structured file naming, and built-in GPS tagging. KoboToolbox is purpose-built for field data collection by non-specialists and handles the exact use case of volunteers capturing images with structured metadata in outdoor settings where connectivity may be unreliable.

3.1.1 Form Design

The collection form should capture the following fields per image submission:

Field	Type	Required	Notes
Volunteer ID	Text / dropdown	Yes	Pre-assigned ID (e.g., S001 for student, A001 for adult). Dropdown prefilled with active volunteer list.
Photo	Image attachment	Yes	Max 10 MB. JPEG or PNG. One image per submission for simplicity.
Farm location	Select one	Yes	Dropdown of participating farm names (not free text).
Plot / area	Text	No	Optional field or row identifier per farm protocol.
Capture conditions	Select multiple	No	Checkboxes: sunny, overcast, wet, dry, morning, afternoon. Aids later filtering.
Notes	Text	No	Freeform. Volunteer observations.
GPS coordinates	Auto-captured	Yes	KoboCollect captures this automatically if location services are enabled.
Timestamp	Auto-captured	Yes	Automatic.

The form should be tested internally during Step B (Weeks 3–4) with 1–2 people before volunteer onboarding.

3.1.2 Volunteer Experience

Volunteers download the KoboCollect app (Android) or use the web form (iOS and desktop). They log in with a shared project link or credentials provided during training. In the field, they open the form, take or attach a photo, fill the short metadata fields, and submit. If they have no signal, submissions queue locally and sync when connectivity returns.

Training requirement: 5–10 minutes during the onboarding session (Step C). A one-page quick reference card should be provided, covering: install app, open form, take photo, submit.

3.1.3 Data Export and Ingestion Pipeline

KoboToolbox provides a REST API and manual CSV/media export. For a pilot of this size (3,000–6,000 images), a semi-automated approach is sufficient:

- Weekly manual export of submissions (CSV + media bundle) by the technical support person.
- Images renamed according to a consistent naming convention: {farm}_{volunteerID}_{date}_{sequence}.jpg.
- Renamed images uploaded to the central Google Drive folder structure (see Section 4).
- CSV metadata appended to a master tracking spreadsheet.

If the volume warrants it, a simple Python script can automate the KoboToolbox API export and renaming. This script should be written during Step B and tested before volunteer data starts flowing.

3.1.4 Fallback: Google Forms + Drive

If KoboToolbox proves problematic (e.g., school IT restrictions on app installation), fall back to Google Forms with image upload fields. This loses offline capability and GPS auto-capture, but remains functional. In this case, GPS should be captured manually via a separate phone screenshot or a simple GPS app, and attached as a second image or noted in the text field.

4. Image Storage and Management

4.1 Primary Storage: Google Drive

For a pilot of this scale, Google Drive (via an institutional Google Workspace account or a dedicated free Google account) provides sufficient storage, access control, and simplicity. The 15 GB free tier accommodates approximately 5,000–7,000 compressed JPEG images at 2–3 MB each.

If an institutional Workspace account is available (e.g., through a participating university or school), use it for larger storage quotas and better access management.

4.1.1 Folder Structure

The Drive should be organised as follows:

Folder Path	Contents	Access
/Pilot01/	Root folder	Shared with core team
/Pilot01/Raw_Uploads/	Unprocessed images as exported from KoboToolbox, organised by export date	Technical support only
/Pilot01/Processed/{FarmName}/	Renamed, validated images ready for annotation, one subfolder per farm	Core team + Label Studio ingestion
/Pilot01/Gold_Standards/	Expert-labelled reference images (not shared with volunteers)	Experts + technical support only
/Pilot01/Exports/	Annotation exports, analysis outputs, model checkpoints	Core team
/Pilot01/Protocols/	Protocol documents, training materials, quick reference cards	All participants (view only)
/Pilot01/Reports/	Weekly summaries, final pilot report drafts	Core team

4.1.2 Backup

The technical support person should maintain a local backup (external drive or second cloud location) of the Processed folder, updated weekly. At 6,000 images this is approximately 12–18 GB, easily handled by any external drive or a free Backblaze B2 bucket (10 GB free).

4.1.3 Alternative: Cloud Object Storage

If the team prefers programmatic access (e.g., for scripted ingestion into Label Studio), a cloud object storage bucket is a reasonable alternative. Backblaze B2 offers 10 GB free; Google Cloud Storage and AWS S3 have free-tier allowances. For this pilot, the added complexity is likely unnecessary unless the technical team is already comfortable with these tools.

5. Annotation and Labelling

5.1 Primary Tool: Label Studio Community Edition

What it is: Label Studio is an open-source, self-hosted data labelling platform. The Community Edition is free and supports image classification, bounding boxes, polygons, and custom labelling interfaces. It runs as a Python web application and can be accessed via any modern browser.

Why Label Studio: It is the most widely adopted open-source annotation tool with active development, good documentation, and a straightforward setup. It supports the label taxonomy structure needed for this pilot (species classes, functional groups, “Other weed,” “Unclassifiable”), handles multi-user workflows, and exports in standard formats (JSON, CSV, COCO, Pascal VOC).

5.1.1 Deployment

Label Studio should be deployed on a single small cloud server or virtual machine. Recommended setup:

Component	Specification	Estimated Cost
Cloud VM	2 vCPU, 4 GB RAM, 50 GB SSD (e.g., Hetzner CX22, DigitalOcean Basic Droplet, or AWS Lightsail)	€5–€15/month
Operating system	Ubuntu 22.04 LTS	Free
Label Studio	Community Edition, installed via pip or Docker	Free
Reverse proxy	Caddy or Nginx with Let’s Encrypt TLS	Free
Domain	Subdomain of an existing domain (e.g., label.abundancefederation.org) or a free subdomain service	Free–€10/year
Database	SQLite (default, sufficient for this scale) or PostgreSQL if team prefers	Free

Deployment steps (estimated time: 2–4 hours):

- Provision VM and point DNS.
- Install Docker or Python environment.
- Run Label Studio (docker run -p 8080:8080 heartexlabs/label-studio:latest or pip install label-studio && label-studio start).
- Configure Caddy/Nginx as reverse proxy with automatic HTTPS.
- Create admin account, configure project.
- Test access from mobile browser to confirm volunteer usability.

5.1.2 Project Configuration

A single Label Studio project is sufficient for this pilot. Configuration details:

Labelling interface: Image classification (single-label). The volunteer sees one image and selects from the defined taxonomy. If bounding boxes are also desired (e.g., marking weed location within the image), the interface should combine classification with a bounding box tool. The expert group should decide this during protocol design (Weeks 1–3). Start with classification-only to keep volunteer training simple; add localisation in a future pilot if needed.

Label taxonomy (to be finalised by experts, illustrative example):

- **Species classes:** Fat Hen, Charlock, Chickweed, Redshank, Creeping Thistle, Docks (grouped), Annual Meadow Grass
- **Functional group:** Broadleaf weed (unspecified), Grass weed (unspecified)
- **Non-species outcomes:** Other weed (identifiable but not in target list), Unclassifiable (insufficient visual information), Not a weed / Crop, No plant visible

The taxonomy is configured in Label Studio's labelling interface XML. Example:

```
<View>    <Image name="image" value="$image"/>    <Choices name="label"
toName="image" choice="single-column">    <Choice value="Fat Hen"/>    <Choice
value="Charlock"/>    ...    <Choice value="Other weed"/>    <Choice
value="Unclassifiable"/>    </Choices> </View>
```

5.1.3 User Management

Label Studio supports user accounts with email/password login. For this pilot:

- Each volunteer gets an individual account (essential for tracking per-person quality metrics).
- Accounts are created in bulk by the technical support person before onboarding (Step C).
- Volunteers log in via mobile or desktop browser. No app installation required.
- Roles: annotator (Level 2 volunteers), reviewer (Level 3), admin (core team and experts).

5.1.4 Image Ingestion

Images from the Processed Google Drive folder are imported into Label Studio. Two approaches:

- **Manual upload:** Zip the batch of processed images and upload via the Label Studio web interface. Sufficient for weekly batches of a few hundred images.
- **Cloud storage sync:** If using cloud object storage (S3/GCS/B2), Label Studio can connect directly and sync images. This is more elegant but adds configuration complexity. Only pursue if the team is comfortable with it.

Each batch import should be logged in the master tracking spreadsheet with: date, number of images, source farm, and any issues noted.

5.1.5 Annotation Workflow

The workflow maps directly to the pilot's operational flow:

- **Round 1 – Annotation (Step E):** Level 2 volunteers label images. Each image is assigned to one annotator. Label Studio's task distribution can be configured to auto-assign or volunteers can claim tasks from a queue.

- **Round 2 – Peer review (Step E):** A second volunteer (ideally from the other cohort, per the cross-cohort review design) reviews the annotation. In Label Studio, this is handled by duplicating the task set into a review project or using the review stream feature. The reviewer sees the image and the original label, and either confirms, corrects, or flags for expert review.
- **Round 3 – Expert audit (Step F):** Experts review a statistically significant sample. Flagged items and a random sample are routed to experts via Label Studio’s admin view or a filtered export.

5.1.6 Volunteer Experience

Volunteers open a browser (phone or laptop), log in to Label Studio, and see their task queue. They click an image, select a label from the dropdown, and submit. The interface is clean and mobile-friendly. A typical annotation takes 15–45 seconds per image once the volunteer is trained.

Training requirement: 10–15 minutes during onboarding (Step C), covering: how to log in, how to read the image, how to select a label, when to choose “Unclassifiable,” and how to submit. Provide a short video walkthrough (2–3 minutes) and a one-page reference card.

6. Quality Assurance and Gold-Standard Management

This is the most operationally critical area of the tooling stack. The pilot's success depends on measuring annotation quality programmatically, without requiring experts to review every item. The tooling must support gold-standard embedding, per-annotator accuracy tracking, and inter-annotator agreement metrics.

6.1 Gold-Standard Item Management

200–300 expert-labelled gold-standard images are embedded invisibly into annotation and review workflows.

Implementation:

- Gold-standard images are stored separately in the Gold_Standards Drive folder, with expert labels recorded in a CSV file (image filename, expert label, difficulty category, edge-case flag).
- Gold items are imported into Label Studio alongside regular images. They are indistinguishable from normal tasks in the volunteer's view.
- A Python tracking script (run weekly by the technical support person) compares volunteer labels on gold items against expert labels and computes per-volunteer accuracy.

6.1.1 Gold Tracking Script

A Python script (gold_tracker.py) should be written during Step A/B and should:

- Export all annotations from Label Studio via the API (JSON export).
- Filter annotations where the image filename matches the gold-standard list.
- For each volunteer, compute: total gold items annotated, exact-match accuracy, confusion matrix against expert labels, correct routing rate for ambiguous items (i.e., did they correctly choose “Other weed” or “Unclassifiable” when the expert did).
- Output a summary CSV and a short human-readable report (per-volunteer scores, flagged volunteers below threshold, overall accuracy).
- This script is run weekly. Results are shared with the pilot coordinator and discussed in the weekly team check-in.

Estimated development time: 4–8 hours for a competent Python developer familiar with Label Studio's API.

6.2 Inter-Annotator Agreement

Where images receive both an annotation (Round 1) and a peer review (Round 2), agreement can be measured. A second script (agreement_tracker.py) should:

- Match Round 1 and Round 2 labels for each image.
- Compute overall agreement rate, Cohen's kappa (or equivalent), and per-class agreement.
- Identify images with disagreement for potential expert triage.
- Output a summary report.

Estimated development time: 2–4 hours, building on the gold tracker script.

6.3 Volunteer Quality Dashboard

Rather than building a custom web dashboard, quality metrics should be maintained in a shared Google Sheet updated weekly by the tracking scripts. Columns:

Column	Description
Volunteer ID	Unique identifier
Cohort	Student / Adult
Current level	0–4 per the progression table
Total annotations	Cumulative count
Gold items seen	Count of gold-standard items annotated
Gold accuracy	% exact match with expert labels
Ambiguity routing	% correct use of Other/Unclassifiable on ambiguous gold items
Agreement rate	% agreement with peer reviewer (where applicable)
Status	Active / Flagged / Paused / Retrained
Notes	Free text for coordinator observations

This sheet serves as the single source of truth for volunteer quality management and feeds into the final pilot report.

7. Analysis and ML Sanity Checking

The pilot does not aim to train a production model. The ML component exists solely to verify that the dataset is usable: that a simple model trained on the labelled data performs meaningfully better than random chance and fails in understandable ways.

7.1 Compute Environment

Google Colab (free tier) is sufficient for the analysis and model training required. The free tier provides intermittent GPU access (typically a T4), which is adequate for training small image classifiers on a dataset of 3,000–6,000 images.

Alternative: If a participating university offers access to a compute cluster or GPU server, use that. If neither is available, a modern laptop with 16 GB RAM can train small models (ResNet-18, MobileNet) on CPU in reasonable time for a dataset this size.

7.1.1 Analysis Notebooks

The data science lead should prepare the following Colab/Jupyter notebooks during Steps A–B, ready to run on real data from Step F onward:

Notebook 1: Dataset Overview

- Class distribution (bar chart of label frequencies).
- Per-farm image counts.
- Per-volunteer contribution counts.
- Image quality summary (resolution distribution, file size distribution).
- Proportion of images routed to Other/Unclassifiable.

Notebook 2: Quality Metrics

- Gold-standard accuracy across all annotators (aggregate and per-person).
- Inter-annotator agreement (overall and per-class).
- Confusion matrix: annotator labels vs. expert labels on gold items.
- Drift analysis: does accuracy change over time (early vs. late in the annotation period)?

Notebook 3: ML Baseline

- Train/validation/test split (stratified by class, ensuring gold items are in the test set for direct comparison).
- Fine-tune a pre-trained model (e.g., ResNet-18 or MobileNetV2 from torchvision or timm) on the training set.
- Report: accuracy, per-class precision/recall/F1, confusion matrix.
- Error analysis: what does the model get wrong? Are failures attributable to protocol gaps, visual ambiguity, class imbalance, or annotation error?
- Compare model performance on expert-labelled gold items vs. volunteer-labelled items.

Estimated development time for all three notebooks: 8–16 hours total, assuming familiarity with PyTorch/fastai and standard image classification workflows.

7.2 Libraries and Dependencies

The following Python libraries should be used (all available in Colab by default or via pip):

Library	Purpose
pandas, numpy	Data manipulation and analysis
matplotlib, seaborn	Visualisation and charting
scikit-learn	Metrics (confusion matrix, kappa, classification report), train/test splits
Pillow (PIL)	Image loading and basic validation
torch, torchvision (or fastai)	Model training and fine-tuning
timm	Pre-trained model zoo (optional, for broader model selection)
label-studio-sdk	Programmatic export from Label Studio API

8. Coordination and Documentation

The coordination layer is deliberately lightweight. The pilot involves a small core team (3–5 people), a small number of experts (2–3), and a manageable number of volunteers (20–60 across both cohorts). Heavy project management tooling would be overhead.

8.1 Google Workspace

Use Google Workspace (Docs, Sheets, Slides, Meet, Drive) as the primary coordination suite. If an institutional account is available, use it; otherwise, free Google accounts are sufficient.

Tool	Use
Google Docs	Protocol documents (v0.1, v0.2, final), training guides, quick-reference cards, pilot report drafts, meeting notes
Google Sheets	Master data tracker (images collected, by farm, by volunteer), volunteer quality dashboard (Section 6.3), weekly status summary, budget tracker
Google Slides	Volunteer training presentation, expert briefing deck
Google Meet	Weekly 30-minute core team check-in, fortnightly expert sync, ad-hoc troubleshooting calls
Google Drive	Central file storage (Section 4)

8.2 Communication

For day-to-day communication between the core team:

- **WhatsApp group or Signal group:** Free, familiar, works on all phones. One group for the core team. A separate group for volunteers is optional but useful for quick announcements, photo sharing, and morale.
- **Email:** For formal communications with schools, farms, experts, and any external partners.

Not recommended for this pilot: Slack, Discord, Microsoft Teams, or other dedicated platforms. They add adoption friction for a 12-week project with a small team. If the team is already using Slack or Teams, that's fine, but do not introduce a new platform solely for this pilot.

8.3 Task Tracking

A shared Google Sheet with a simple task list is sufficient for tracking implementation work during the setup phase (Weeks 1–4). Columns: task description, owner, status (not started / in progress / done), due date, notes. No need for Jira, Asana, or Trello at this scale.

9. Cost Estimate

The following table provides a realistic cost estimate for the full 12-week pilot. The aim is to stay within €500–€1,500 total.

Item	Tool / Service	Cost	Notes
Data collection platform	KoboToolbox (hosted)	€0	Free for non-commercial use
Cloud VM for Label Studio	Hetzner CX22 or DigitalOcean Basic (3 months)	€15–€45	~€5–€15/month
Domain / subdomain	Existing domain or free service	€0–€10	One-time or annual
Google Workspace	Free tier or institutional	€0	Assumes institutional or free accounts
Google Colab Pro (optional)	Monthly subscription if free tier insufficient	€0–€30	€10/month, only if needed
Image backup storage	Backblaze B2 free tier or external drive	€0–€50	External drive if preferred
Printed materials	Quick reference cards, consent forms	€20–€50	Printing costs
Contingency	Unexpected tool costs or upgrades	€50–€150	Buffer
Total estimated range		€85–€335	Well within budget target

10. Implementation Timeline

This timeline needs to be agreed with volunteers, it is a holding space to ensure a timeline is agreed but given this is the first dataset project it is understood that time estimates will be hard to predict with accuracy.

Week	Pilot Phase	Tooling Tasks
1	Protocol design begins	Provision cloud VM. Install and configure Label Studio. Set up Google Drive folder structure. Set up KoboToolbox account and draft collection form. Begin writing gold tracker and agreement scripts.
2	Protocol design continues	Finalise KoboToolbox form based on protocol draft. Configure Label Studio labelling interface with draft taxonomy. Create volunteer account templates. Test full data flow: KoboToolbox → export → rename → Drive → Label Studio import.
3–4	Internal test (Step B)	1–2 internal testers use KoboToolbox to collect ~100 images. Images processed through full pipeline. Test Label Studio annotation workflow end to end. Test gold-standard embedding. Refine scripts and fix issues. Prepare training materials (slides, reference cards, video walkthrough).
4–5	Volunteer onboarding (Step C)	Create volunteer accounts in Label Studio. Ensure mobile access works reliably. Support training sessions with live demo. Distribute reference cards.
5–6	Data collection (Step D)	Monitor KoboToolbox submissions daily. Run weekly export and processing. Upload processed images to Drive and Label Studio. Track volume in master spreadsheet.
6–8	Annotation & peer review (Step E)	Monitor annotation progress in Label Studio. Run gold tracker script weekly. Run agreement tracker weekly. Flag volunteers below threshold for coordinator action. Manage task assignment for peer review round.
8–10	Expert audit & ML (Step F)	Export annotations for expert review. Run Dataset Overview notebook. Run Quality Metrics notebook. Train baseline ML model (Notebook 3). Share results with experts and coordinator.
10–12	Synthesis & reporting (Step G)	Assemble final dataset export (images + labels + metadata). Write dataset card. Compile quality metrics into pilot report. Archive all data and scripts. Document lessons learned on tooling.

11. Tooling Risks and Mitigations

Risk	Likelihood	Impact	Mitigation
School IT blocks KoboCollect app installation	Medium	Medium	Fall back to KoboToolbox web form (no app needed) or Google Forms. Test with school IT department during Week 2.
Label Studio VM goes down during annotation period	Low	High	Daily automated backups of Label Studio database (SQLite file). VM can be re-provisioned and restored within hours. Keep backup on Drive.
Volunteers struggle with Label Studio on mobile	Medium	Medium	Test thoroughly on both iOS and Android browsers during Week 2–3. Simplify labelling interface if needed (fewer options per screen). Provide fallback: volunteers can annotate on laptops during supervised sessions.
Image volume exceeds Google Drive free tier	Low	Low	Upgrade to Google One (€2/month for 100 GB) or use institutional Workspace account. Not a meaningful cost.
KoboToolbox hosted service has downtime	Low	Medium	Data collection can pause for a day without pilot impact. If persistent, switch to Google Forms fallback.
Gold-standard items are too few or poorly distributed across classes	Medium	High	Expert creates gold items during Week 1–2 with deliberate coverage of all classes including edge cases. Aim for at least 15–20 gold items per class.
Tracking scripts produce incorrect metrics	Medium	Medium	Validate scripts against a small hand-checked sample during internal test (Week 3–4). Include unit tests for core calculation functions.

12. Post-Pilot: What Happens to the Tooling

All tooling in this stack is intentionally disposable. After the pilot, the team should:

- Export and archive the complete dataset (images, annotations, metadata, gold standards) to durable storage (e.g., university research archive, institutional Drive, or a cloud bucket with long-term retention).
- Archive all scripts, notebooks, protocol documents, and tracking sheets alongside the dataset.
- Write a tooling lessons-learned section in the pilot report, covering: what worked, what was friction, what should change for the next pilot.
- Decommission the Label Studio VM (cancel the cloud instance) to avoid ongoing costs.
- Feed tooling lessons into the design of the federation's long-term open-source infrastructure stack.

The explicit goal is that Pilot 01's tooling experience informs the design of a purpose-built, scalable platform for future data collection campaigns. No tool chosen here should be treated as a permanent commitment.